
University of Utah

Fall 2025

MATH 3220-002

Midterm 1 Questions

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September 12, 2025, 9:40 AM - 10:30 AM

Surname:

First Name:

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KEY

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make sure to read and
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document, distributed
separately.**

1. [40 points] Let $f : V \rightarrow W$ be an arbitrary function between vector spaces.

(a) [20 points] **Prove or disprove the following statement:**

If f is linear, then its graph is a linear subspace of $V \times W$.

Let $(v_1, w_1), (v_2, w_2) \in \text{graph}(f)$, $c \in \mathbb{R}$.

$$\Rightarrow \boxed{w_1 = f(v_1) \text{ and } w_2 = f(v_2)}$$

$$\Rightarrow c(v_1, w_1) + (v_2, w_2) = (cv_1 + v_2, cw_1 + w_2)$$

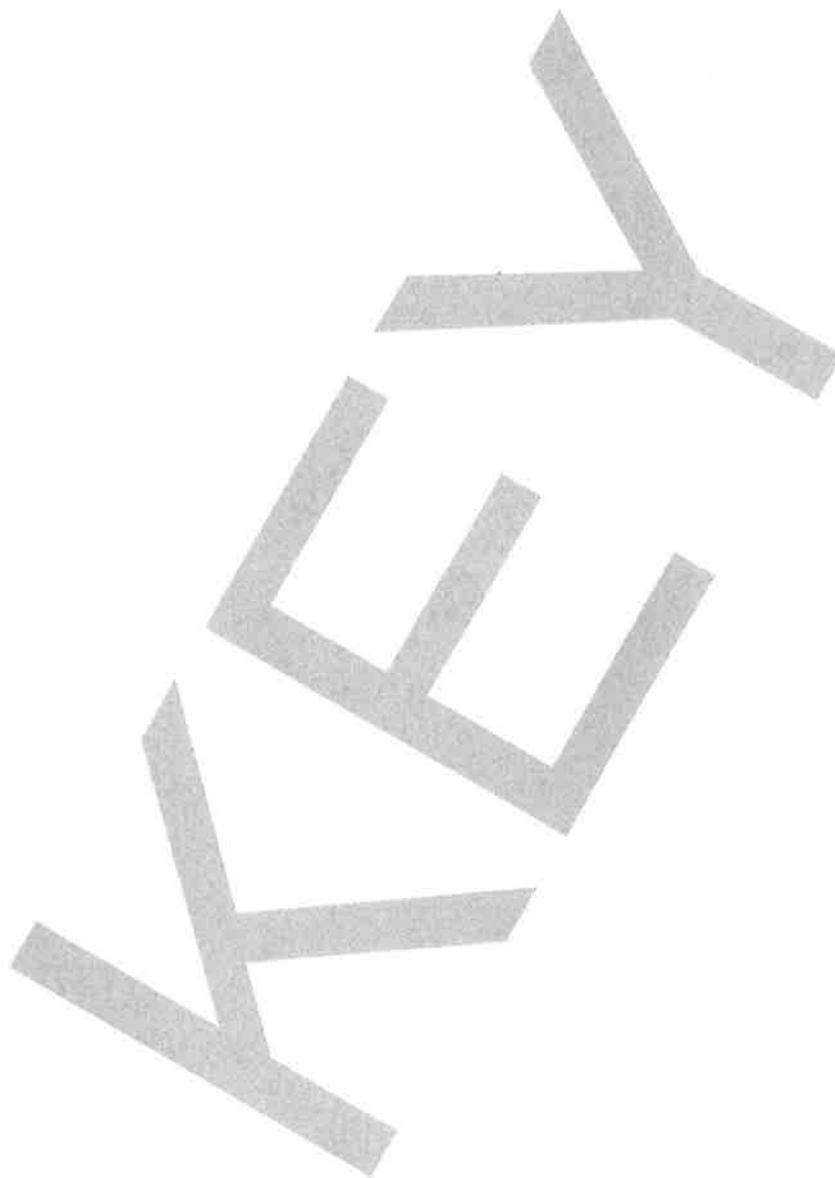
$$= (cv_1 + v_2, c f(v_1) + f(v_2))$$

$$= (cv_1 + v_2, f(cv_1 + v_2)) \in \text{graph}(f).$$

$\uparrow$
 f is linear

$\square$

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(b) [20 points] Prove or disprove the following statement:

If the graph of f is linear, then f is linear.

$$\text{Let } c \in \mathbb{R}, v_1, v_2 \in V.$$

$$\Rightarrow (v_1, f(v_1)), (v_2, f(v_2)) \in \text{graph}(f)$$

Since $\text{graph}(f)$ is linear,

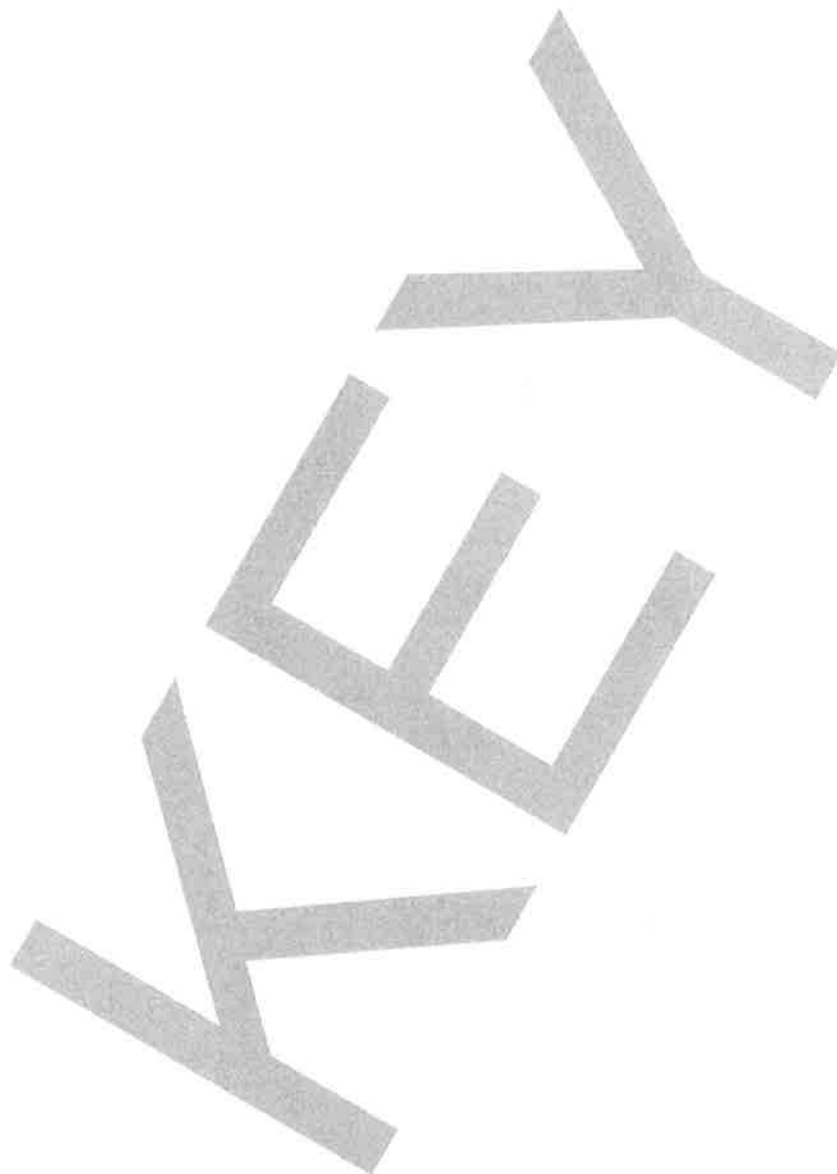
$$c(v_1, f(v_1)) + (v_2, f(v_2)) \in \text{graph}(f)$$

$$\Rightarrow (cv_1 + v_2, cf(v_1) + f(v_2)) \in \text{graph}(f)$$

$$\Rightarrow cf(v_1) + f(v_2) = f(cv_1 + v_2) \quad \checkmark$$

□.

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2. [43 points] Let $\mathcal{M}$ be the vector space of $n \times n$ matrices with real entries.

(a) [40 points] **Prove or disprove the following statement:**

For any $A, B \in \mathcal{M}$, the operator norm of AB is less than or equal to the product of the operator norms of A and B :

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}.$$

Let $\alpha, \beta \geq 0$ be such that
 $\forall v, w \in \mathbb{R}^n: |Av| \leq \alpha |v|$
 $|Bw| \leq \beta |w|.$

Then $\forall w \in \mathbb{R}^n:$

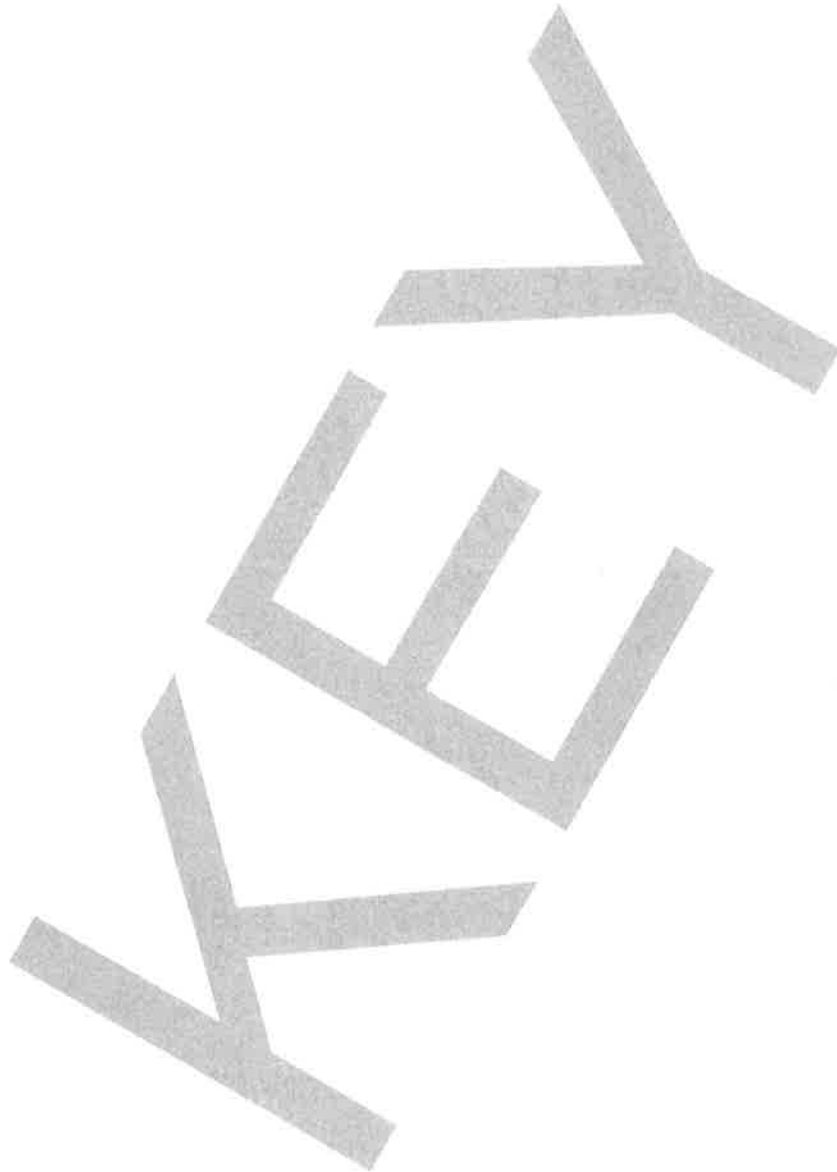
$$\begin{aligned} |(AB)w| &= |A(Bw)| \leq \alpha |Bw| \\ &\leq \alpha \beta |w| \end{aligned}$$

$$\Rightarrow \|AB\|_{\text{op}} \leq \alpha \beta.$$

Optimizing over α and β gives:

$$\|AB\|_{\text{op}} \leq \|A\|_{\text{op}} \|B\|_{\text{op}}. \quad \square.$$

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(b) [3 points] Prove or disprove the following statement:

If $\|\bullet\|$ is an arbitrary norm on $\mathcal{M}$, then for any $A, B \in \mathcal{M}$

$$\|AB\| \leq \|A\|\|B\|.$$

Consider the entrywise ℓ^∞ norm,
say for $n=2$:

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \Rightarrow \|A\|_{\ell^\infty} = \max\{|a|, |b|, |c|, |d|\}.$$

$$\text{Put } A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

$$\Rightarrow AB = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}.$$

$$\Rightarrow \|A\|_{\ell^\infty} \|B\|_{\ell^\infty} = 1 \cdot 1 = 1$$

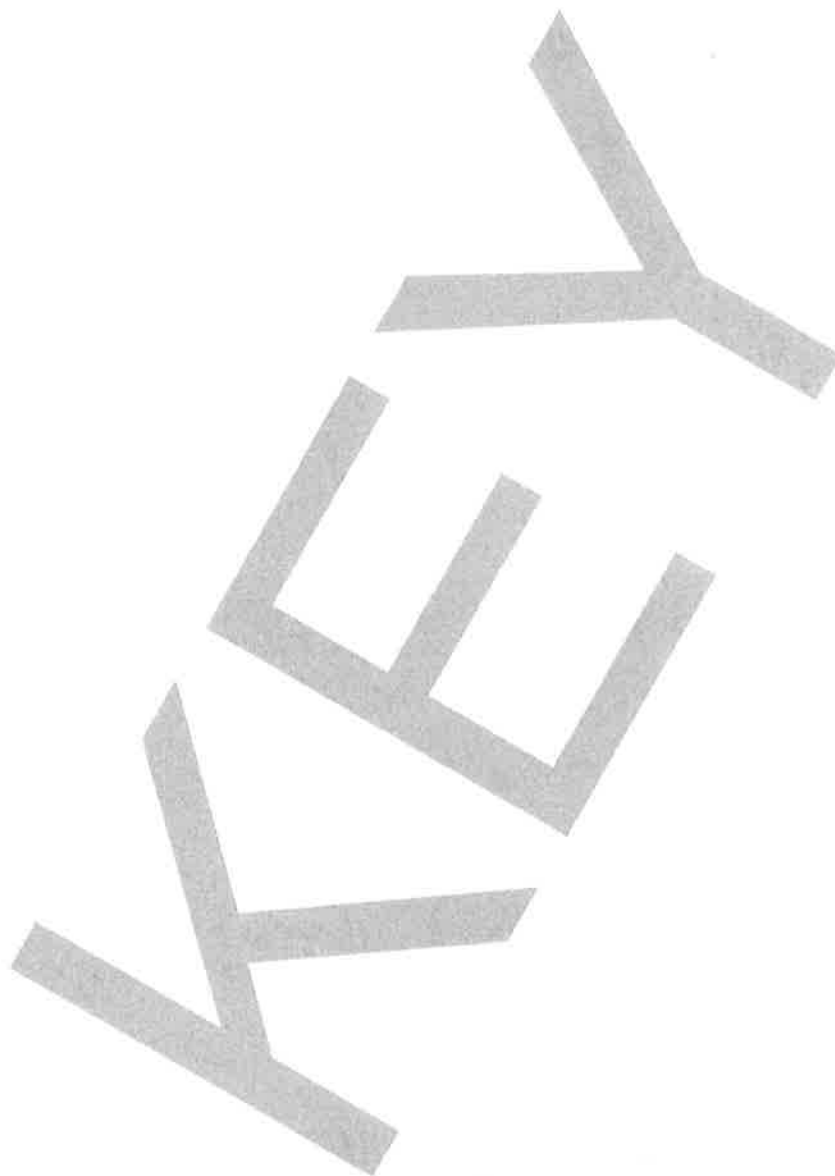
$$\|AB\|_{\ell^\infty} = 2 > \|A\|_{\ell^\infty} \|B\|_{\ell^\infty}.$$

Generalize to arbitrary n :

$$A = \begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{n1} & \dots & a_{nn} \end{pmatrix} \Rightarrow \|A\|_{\ell^\infty} = \max_{1 \leq i, j \leq n} |a_{ij}|$$

$$A = \begin{pmatrix} 1 & 0 & \dots & 0 & 1 \\ & \ddots & & & \vdots \\ & & 1 & & 0 \\ & & & \ddots & \\ 0 & & & & 1 \end{pmatrix}, B = A^T.$$

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3. [13 points] **Prove or disprove the following statement:** Let V be a vector space and $\langle \bullet, \bullet \rangle$ be an inner product on V . Then the inner product $\langle \bullet, \bullet \rangle$, thought of as a function $V \times V \rightarrow \mathbb{R}$, is continuous.

Let $x_i \rightarrow x_\infty, y_i \rightarrow y_\infty$ in V

(relative to the metric d defined by

$$d(v_1, v_2) = \sqrt{\langle v_1 - v_2, v_1 - v_2 \rangle}).$$

$$|\langle x_n, y_n \rangle - \langle x_\infty, y_\infty \rangle|$$

$$= |\langle x_n, y_n \rangle - \langle x_n, y_\infty \rangle + \langle x_n, y_\infty \rangle - \langle x_\infty, y_\infty \rangle|$$

$$\stackrel{\text{TI}}{\leq} |\langle x_n, y_n \rangle - \langle x_n, y_\infty \rangle| + |\langle x_n, y_\infty \rangle - \langle x_\infty, y_\infty \rangle|$$

$$= |\langle x_n, y_n - y_\infty \rangle| + |\langle x_n - x_\infty, y_\infty \rangle|$$

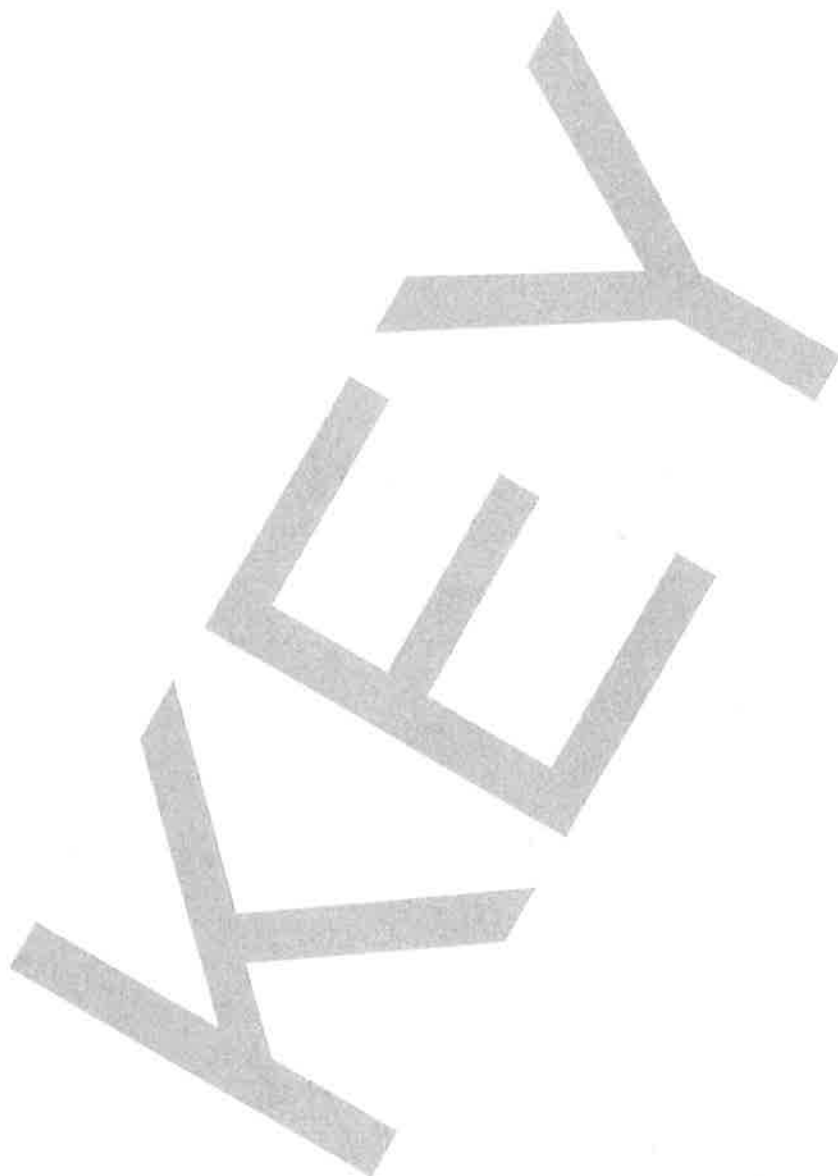
linearity

$$\stackrel{\text{CBS}}{\leq} \underbrace{|x_n|}_{\rightarrow |x_\infty|} \underbrace{\|y_n - y_\infty\|}_{\rightarrow 0} + \underbrace{|x_n - x_\infty|}_{\rightarrow 0} \underbrace{\|y_\infty\|}_{< \infty} \xrightarrow{n \rightarrow \infty} 0$$

(*)

$$(*) : \underbrace{||x_n| - |x_\infty||}_{\text{Reverse TI}} \leq |x_n - x_\infty| \xrightarrow{n \rightarrow \infty} 0$$

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4. [4 point] Let d be a positive integer and $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$. The **mean** $\mu(x)$ and **standard deviation** $\sigma(x)$ of x are defined as follows:

$$\mu(x) = \frac{1}{d} \sum_{k=1}^d x_k, \quad \sigma(x) = \left(\frac{1}{d} \sum_{k=1}^d |x_k - \mu(x)|^2 \right)^{1/2}.$$

- (a) [1 point] **Prove or disprove** the following statement:
For any $k \in \{1, 2, \dots, d\}$,

$$|x_k - \mu(x)| \leq \sqrt{d-1} \sigma(x).$$

Fix k_0 . $d\mu = \sum_{k=1}^d x_k \Rightarrow \sum_{k=1}^d (x_k - \mu) = 0$

$$\Rightarrow x_{k_0} - \mu = \sum_{k \neq k_0} (x_k - \mu).$$

Define $\tilde{x}_{k_0} = (x_1, x_2, \dots, x_{k_0-1}, x_{k_0+1}, \dots, x_d) \in \mathbb{R}^{d-1}$

$$\mathbf{1}_{d-1} = (1, 1, \dots, 1, 1, \dots, 1) \in \mathbb{R}^{d-1}.$$

$$\Rightarrow \left| \sum_{k \neq k_0} (x_k - \mu) \right| = \left| \langle \tilde{x}_{k_0} - \mu \mathbf{1}_{d-1}, \mathbf{1}_{d-1} \rangle \right|$$

$$\stackrel{(*)}{\leq}_{CBS} \left| \tilde{x}_{k_0} - \mu \mathbf{1}_{d-1} \right| \cdot \left| \mathbf{1}_{d-1} \right|.$$

$$\left| \tilde{x}_{k_0} - \mu \mathbf{1}_{d-1} \right| = \left(\sum_{k \neq k_0} |x_k - \mu|^2 \right)^{1/2}$$

$$= \left[\left(\sum_{k=1}^d |x_k - \mu|^2 \right) - |x_{k_0} - \mu|^2 \right]^{1/2} \quad \text{continued} \rightarrow$$

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$$\begin{aligned} |\mathbf{1}_{d-1}| &= \sqrt{d-1} \\ \Rightarrow |x_{k_0} - \mu|^2 &\leq |\tilde{x}_{k_0} - \mu \mathbf{1}_{d-1}|^2 |\mathbf{1}_{d-1}|^2 \\ &= \left[d \sigma^2 - |x_{k_0} - \mu|^2 \right] \cdot (d-1) \\ &= d(d-1) \sigma^2 - (d-1) |x_{k_0} - \mu|^2 \\ \Rightarrow d \cdot |x_{k_0} - \mu|^2 &\leq d(d-1) \sigma^2 \\ \Rightarrow |x_{k_0} - \mu| &\leq \sqrt{d-1} \cdot \sigma. \quad \square \end{aligned}$$

(b) [2 points] Prove or disprove the following statement:

Let $k \in \{1, 2, \dots, d\}$. If there is a number $c \in \mathbb{R}$ such that for any $i \in \{1, 2, \dots, d\}$ with $i \neq k$, $x_i = c$, then

$$|x_k - \mu(x)| = \sqrt{d-1} \sigma(x).$$

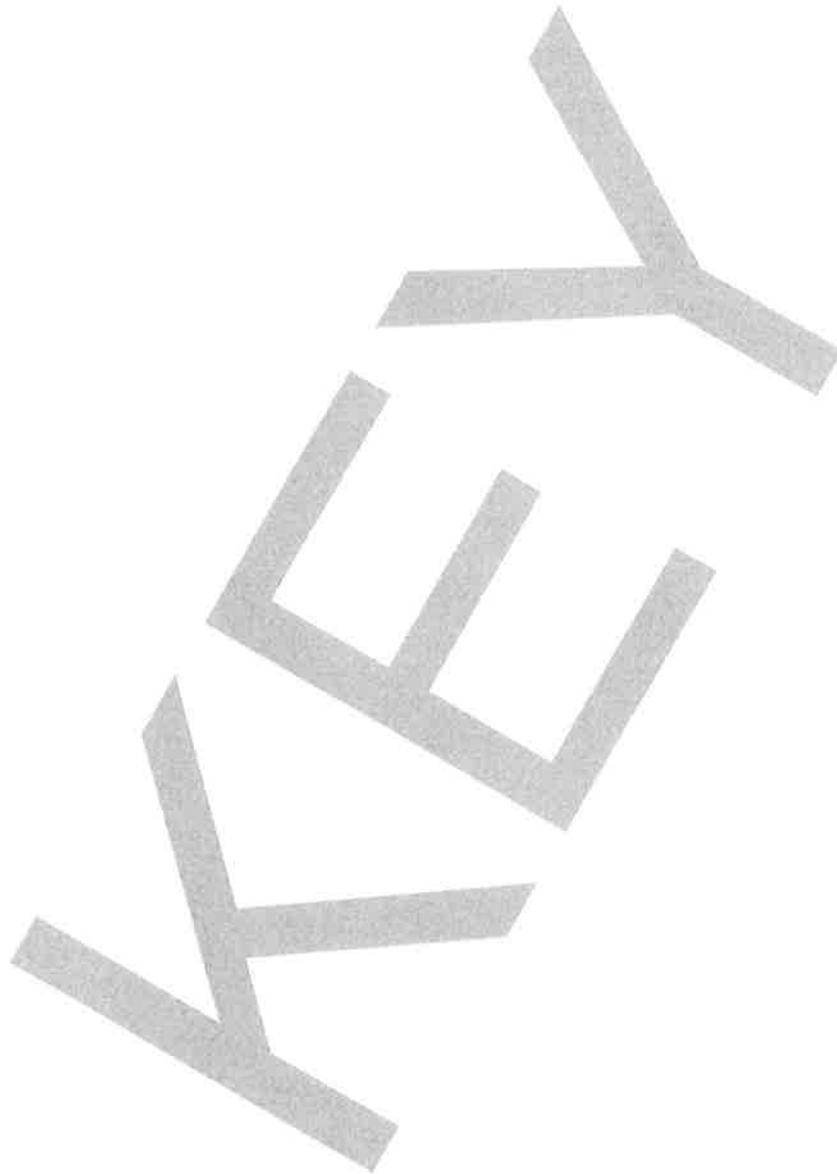
Equality $\Leftrightarrow$ in CBS step (A)

In previous part,

$\tilde{x}_k - \mu \mathbb{I}_{d-1}$ and $\mathbb{I}_{d-1}$
are linearly dependent

$$\Leftrightarrow \tilde{x}_k = c \mathbb{I}_{d-1}. \quad \square$$

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(c) [1 point] **Prove** or disprove the following statement:

Let $k \in \{1, 2, \dots, d\}$. If

$$|x_k - \mu(x)| = \sqrt{d-1} \sigma(x),$$

then there is a number $c \in \mathbb{R}$ such that for any $i \in \{1, 2, \dots, d\}$ with $i \neq k$, $x_i = c$.

See the solution to the previous problem.

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